

Finite extensions of fields

Defⁿ $k \leq L$: field extension

This extension is finite if

$\dim_k(L) < \infty$. In this case, we

set

$$[L:k] = \dim_k(L)$$

"degree of L over k "

Example: $f(x) \in k[x]$

irreducible
of deg d

$$L = k[x]/(f(x))$$

|

k

} finite
extension
of
deg d .

Defⁿ (1) A field extension L/k
"L over k"

is monogenic if $\exists$ irred polynomial
 $f(x) \in k[x]$

and an isomorphism of fields

$$\sigma : k[x]/(f(x)) \xrightarrow{\cong} L$$

$$\text{s.t. } \sigma(a) = a, \quad \forall a \in k$$

Rmk. Monogenic extensions are always finite.

Defⁿ (3) L/k : field extension
 $S \subseteq L$

The subextension generated by S

is the smallest field extension $k(S)/k$
contained in L and containing S

• If $S = \{d\}$, we will just write $k(d)$ instead of $k(\{d\})$.

• If $S = \{d_1, \dots, d_r\}$, we will write $k(d_1, \dots, d_r)$ instead of $k(\{d_1, \dots, d_r\})$.

Observation (1)

If $S = \{d\}$, we can explicitly write

$$k(d) = \left\{ \frac{p(d)}{q(d)} : \begin{array}{l} p(x) \in k[x] \\ q(x) \in k[x] \\ q(d) \neq 0 \end{array} \right\}$$

"rational fn of d"

$$= \{ q(d)^{-1} p(d) : \quad " \quad \}$$

$$\subseteq L$$

Pr.: RHS is a subfield of L

• $1, 0 \in \text{RHS}$ ✓

$$\frac{p_1(x)}{q_1(x)} + \frac{p_2(x)}{q_2(x)} = \frac{p_1(x)q_2(x) + p_2(x)q_1(x)}{q_1(x)q_2(x)}$$

$$\frac{p_1(x)}{q_1(x)} \cdot \frac{p_2(x)}{q_2(x)} = \frac{p_1(x)p_2(x)}{q_1(x)q_2(x)}$$

$$\frac{p(x)}{q(x)} \neq 0 \Rightarrow p(x) \neq 0$$

$$\Rightarrow \left(\frac{p(x)}{q(x)} \right)^{-1} = \frac{q(x)}{p(x)} \quad \checkmark$$

Observation (2) $L/k, \alpha \in L$

$$\begin{array}{ccc} k[x] & \xrightarrow{\text{ev}_\alpha: x \mapsto \alpha} & k[\alpha] \\ \downarrow & \searrow & \parallel \\ k[x]/\ker \text{ev}_\alpha & \xrightarrow{\cong} & \text{im ev}_\alpha \leq L \end{array}$$

There are two possibilities:

(a) $\ker \text{ev}_\alpha = (0)$

"polynomial in α "

$$k[x] \xrightarrow{\cong} k[\alpha] = \left\{ p(\alpha) : p(x) \in k[x] \right\}$$

not a field

$$(h) \quad \text{per } \alpha \mapsto (m_\alpha(x))$$

minimal
polynomial
for α
over k

Where $m_\alpha(x)$ is the unique
irreducible polynomial with
leading coeff. 1 s.t. $m_\alpha(\alpha) = 0$

$$k[x] / (m_\alpha(x)) \xrightarrow{\cong} \underbrace{k[\alpha]}_{\text{field!}}$$

Defⁿ

Case (a):

α is

transcendental
over k

Case (b):

α is

algebraic
over k

Fact (1): e, π are both transcendental
over $\mathbb{Q}$.

Fact (2): Almost every $\alpha \in \mathbb{R}$ is

transcendental over $\mathbb{Q}$.

Example: $\sqrt{2}$: irrational
but algebraic / (4)

Proposition (1)

$L/k, \alpha \in L$

TFAE:

(1) α is algebraic over k

(2) $\ker \text{ev}_\alpha \neq (0)$

(3) $k[\alpha] = k(\alpha)$

(4) $k[\alpha]$ is a field.

(5) $k(\alpha)$ is a finite extension of k .

pf: (1) $\Leftrightarrow$ (2) : By definition

(2) $\Rightarrow$ (3)

(2) holds $\Rightarrow k[\alpha]$ is a field

But we always have $k[\alpha] \subseteq k(\alpha)$

$$\Rightarrow k[\alpha] = k(\alpha).$$

(3) $\Leftrightarrow$ (4) : Immediate

(4) $\Rightarrow$ (2) : When $k[\alpha]$ is a field

that means that $[\alpha \neq 0]$

$$\alpha^{-1} \in k[\alpha]$$

$$\Leftrightarrow \alpha^{-1} = p(\alpha), \quad p(x) \in k[x]$$

$$\Leftrightarrow \alpha p(\alpha) - 1 = 0$$

$\Rightarrow \alpha$ is a zero for $x p(x) - 1$
 $\in k[x]$

0
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$$\Rightarrow x_p(r) - 1 \in \ker \varphi_n$$